6. Ismail ordered a collection of beads. He received 50 beads in all. Out of that 15 beads were brown. Find the percentage of brown beads?
7. Ramu scored 20 out of 25 marks in English, 30 out of 40 marks in Science and 68 out of 80 marks in mathematics. In which subject his percentage of marks is best?
8. Peter requires 50% to pass. If he gets 280 marks and falls short by 20 marks, what would have been the maximum marks of the exam?
9. Kayal scored 225 marks out of 500 in revision test 1 and 265 out of 500 marks in revision test 2. Find the percentage of increase in her score.
10. Roja earned ₹ 18,000 per month. She utilized her salary in the ratio 2:1:3 for education, savings and other expenses respectively. Express her usage of income in percentage.

2.3 Simple Interest

Selvam said that her sister took a loan from the bank to do her higher studies. The loan money that she borrowed from the bank is known as Sum or Principal. The borrower takes some time period to return the money to the bank. To use the money during a particular period of time, the borrower has to pay an additional amount to the bank. This is known as Interest. So to repay the money borrowed, the borrower has to add the principal and the interest.

That is, Amount = Principal+Interest

Interest is generally given in per cent for a period of 1 year per annum. Suppose the bank gives an amount of ₹ 100 at an interest rate of ₹ 8, it is written as 8% per year or per annum or in short 8% p.a. (per annum).



It means on every ₹ 100 borrowed, ₹ 8 is the required interest, to be paid for every one year. To understand this let us consider an example.

Selvam takes a loan of ₹ 10000 at 15% per year as rate of interest. Let us find the interest he has to pay at the end of 1 year.

Sum borrowed = ₹ 10000

Rate of interest = 15 % per year

This means if ₹ 100 is borrowed he has to pay ₹ 15 as interest.

So, for the borrowed amount of ₹ 10000, the interest for one year would be

$$\frac{15}{100} \times 10000 = ₹\,1500$$

So at the end of 1 year, he has to give an amount of = 10000 + 1500 = ₹ 11500.

Now we can write a general relation to find interest for one year. Take P as the principal or sum and r % as Rate per cent per annum. On every ₹ 100 borrowed the interest paid is ₹ r. Therefore, on ₹ P borrowed the interest paid for 1 year would be $\frac{P \times r}{100}$. If the amount

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